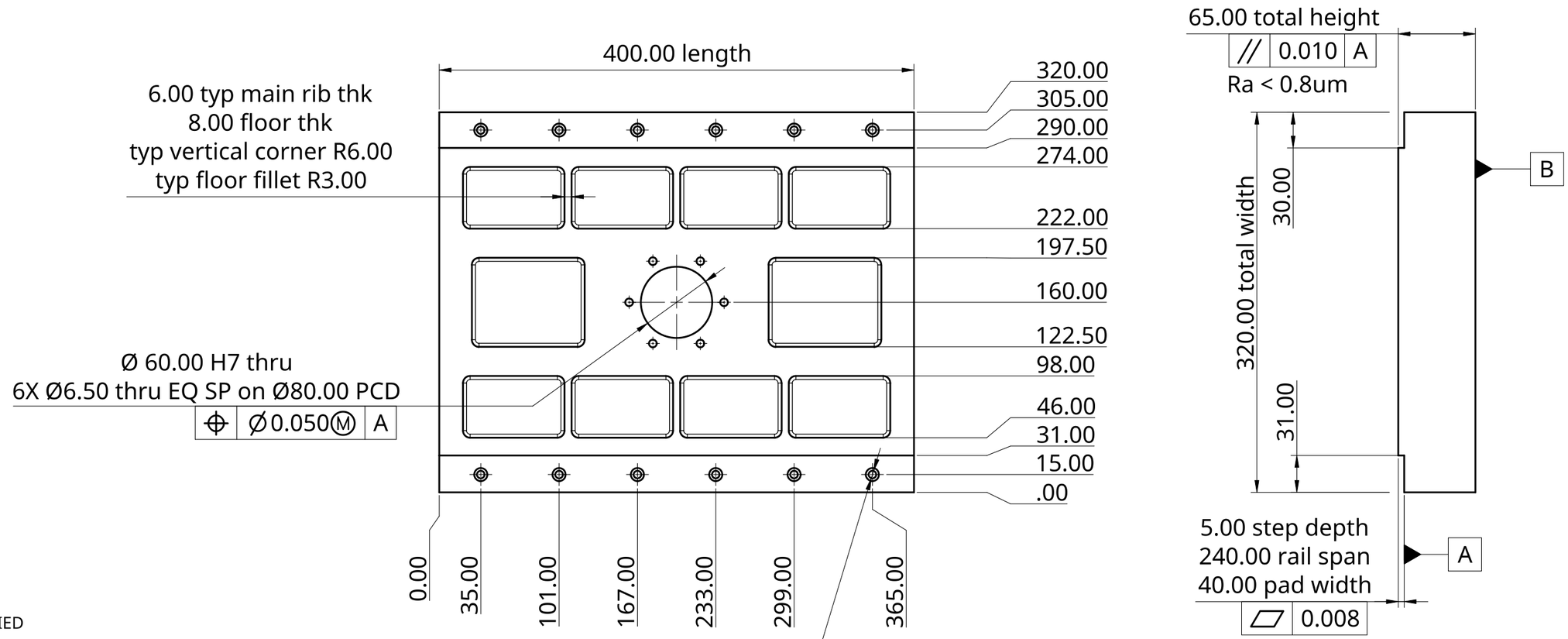
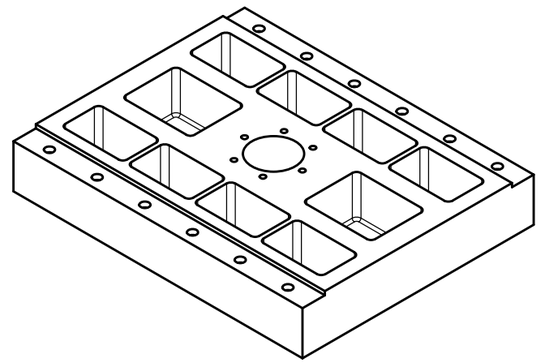




NOTES: UNLESS OTHERWISE SPECIFIED

1. GENERAL & STANDARDS:
   1.1 INTERPRET DRAWING IN ACCORDANCE WITH ASME Y14.5-2018 AND ISO 8015.
   1.2 3D CAD DATASET (W3-CNC-SDL-400_RevA.STEP) GOVERNS NOMINAL GEOMETRY.
   1.3 GENERAL TOLERANCES: LINEAR +/- 0.05 mm, ANGULAR +/- 0.2 DEG (PER ISO 2768-m).
   1.4 THIRD ANGLE PROJECTION APPLIES THROUGHOUT. ALL UNITS IN MILLIMETERS.

2. MATERIAL & BLANK PREPARATION:
   2.1 MATERIAL: HIGH-STRENGTH PRE-STRETCHED ALUMINUM ALLOY 7075-T651 PER ASTM B209 / AMS 4045.
   2.2 MECHANICAL PROPERTIES: YIELD STRENGTH >= 505 MPa, TENSILE STRENGTH >= 570 MPa, DENSITY = 2.81 g/cm^3.
   2.3 RESIDUAL STRESS RELIEF: ULTRASONICALLY STABILIZED PRIOR TO HIGH-SPEED FINISH MACHINING.
   2.4 THEORETICAL CALCULATED FINISHED MASS: 13.85 kg.

3. MANUFACTURING & CNC DFM CONTROLS:
   3.1 1-SETUP 3-AXIS CNC HIGH-SPEED MILLING (SPINDLE >= 18,000 RPM, ZERO LATERAL UNDERCUTS).
   3.2 POCKET CORNER RADIUS R6.0 mm AND FLOOR RADIUS R3.0 mm MANDATORY TO ELIMINATE STRESS CONCENTRATION.
   3.3 MOUNTING DATUM FACES (DATUM A & B) FINISHED VIA PRECISION GROUND/HIGH-SPEED FLY CUTTING.
   3.4 DATUM A GUIDE STEP FLATNESS <= 0.008 mm, Ra <= 0.4 MICRONS.
   3.5 NON-MACHINED POCKET INTERIOR Ra <= 1.6 MICRONS.

4. CLEANROOM HARDENING & SURFACE TREATMENT:
   4.1 HARD ANODIZE COATING PER MIL-A-8625 TYPE III, CLASS 1 (NON-DYED NATURAL).
   4.2 ANODIC COATING THICKNESS: 25 - 30 MICRONS (MICRO-HARDNESS >= 450 HV).
   4.3 HYDROTHERMAL SEALING VIA NICKEL ACETATE BATH (ISO CLASS 3 ZERO-PARTICLE COMPLIANT).
   4.4 MASKING DIRECTIVE: 100% PREVENT OXIDE ACCRETION ON CENTER BORE (DIA 60.00 H7) AND DATUM A.

5. INSPECTION & QUALITY ASSURANCE:
   5.1 CMM 100% FULL-DIMENSIONAL AUDIT ON DATUM A/B PARALLELISM AND PCD HOLE LOCATIONS.
   5.2 CLEANED WITH MULTI-STAGE DEIONIZED ULTRASONIC CAVITATION; PACKED IN ESD-SAFE SEALED BAGS.

UNLESS OTHERWISE SPECIFIED, DIMENSIONS ARE IN MILLIMETERS		NAME	SIGNATURE	DATE
ANGULAR = ± °	DRAWN			2026-10-16
SURFACE FINISH	CHECKED			
DO NOT SCALE DRAWING	APPROVED			
BREAK ALL SHARP EDGES AND REMOVE BURRS				
THIRD ANGLE PROJECTION	MATERIAL: Aluminum Alloy 7075-T651 per ASTM B209 / AMS 4045		FINISH	

TITLE: PRECISION CNC SADDLE WITH DYNAMIC POCKETS

SIZE	DWG NO.	REV.
A3	W3-CNC-SDL-400_REVA	A

SCALE	WEIGHT	SHEET
NTS	13.85 kg	1 of 1